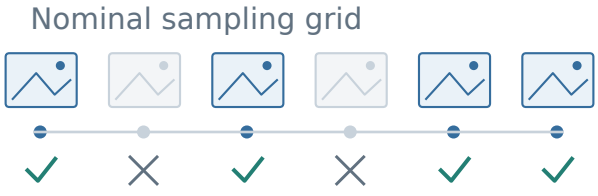
From video samples to Gaussian appearance

a Sample and register



b Gaussian parameters

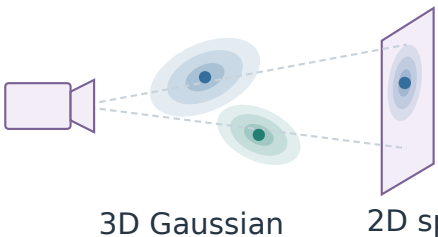
$$s = \exp(\ell_s)$$

$$\alpha = \sigma(\ell_\alpha)$$

SH degree 0–3

$$\Sigma = R(q)\,\mathrm{diag}(s^2)\,R(q)^\top$$

c Project, shade and composite



$$\mathrm{SH}_i(v) = \sum_{\ell,m} \beta_{i,\ell m} Y_{\ell m}(v)$$

SH basis → view color

$$C = \sum_i T_i a_i c_i, \quad T_i = \prod_{j<i} (1 - a_j)$$

Projected alpha · front-to-back order

Raw video feeds Aholo; extracted frames support reference and review.